

# The Interview-Compatible Recommender: A Measuring Instrument for Cold-Start Conversational Recommendation

Thesis Chapter A (v2)

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## Abstract

Before one can study *how* to interview a cold-start user—which questions to ask, in what order—one needs a recommender that can *be* interviewed: a model that folds an arbitrary-length, mixed-channel set of answers into its belief and re-ranks a large catalogue, without per-user retraining and without trading away full-profile accuracy. We argue that this recommender is a *measuring instrument* for the entire elicitation-policy literature, and that its requirements are more demanding than any single published system meets. We state five requirements—**R1** state-of-the-art full-profile accuracy, **R2** arbitrary-length cold-start fold-in, **R3** channel-agnostic input (items, out-of-catalogue concepts, entities, and continuous directions through one interface), **R4** an uncertainty-native belief, and **R5** monotonicity (a truthful answer never lowers ranking quality)—and an acceptance battery (per-build gates **G0–G9** plus one-time certifications **C1–C3**) that operationalises them. A survey of six model families shows the literature is *bifurcated*: the models that hit the full-profile bar (shallow linear/autoencoder recommenders such as EASE [33] and RecVAE [31]) are item-only point estimators, while the models with an uncertainty-native, multi-channel belief (soft-attribute Gaussian elicitation [4], unified item+attribute bandits [17], conjugate critique folds [36]) do not report full-profile accuracy at the R1 bar. We are aware of no published system that occupies the conjunction of all five. The instrument we propose to close this gap is a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian belief layer over its latent, into which every channel folds as a linear observation; three of its input legs—signed/graded values, out-of-catalogue concepts, and continuous direction tokens—are architecturally inexpressible in the item-indicator basis of the R1-bar models. This chapter contributes the requirement framework, the taxonomy and gap analysis, the instrument’s design, and the experimental protocol (a metric-certification bridge to the published ML-20M numbers, then G0/G1/G2 on the ML-25M harness). Our in-house RecVAE realisation sets the bar at full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M protocol, and the graded-native interview tower now *meets* it: a single certified training run reaches 0.3482/0.2462 on the held-out test cohort—a paired-bootstrap tie with the bar—while reproducing the frozen decoder’s ranking exactly at the empty interview, so G0 (full strength preserved) is passed rather than assumed. The remaining per-build gates have all been run on a pre-certification snapshot of the same architecture and pass. The interview-side results (the graded-input curve, the concept channel) share that provenance and are labelled as such throughout; their re-runs on the certified tower are queued.

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# 1 Introduction: the recommender as a measuring instrument

A *cold-start* conversational recommender must elicit taste from a short interview and then rank a large catalogue. Two questions are entangled: *what to ask* (the elicitation policy) and *how to turn answers into recommendations* (the recommender). The elicitation-policy literature is large and active, but almost every comparison in it is confounded: a policy’s apparent value depends on the recommender it drives, and reported gains routinely conflate a better *question-selection strategy* with a better *answer-ingestion model*. Before we can measure elicitation strategies in isolation, we need to fix the recommender—and not just any recommender, but one strong and flexible enough that a good interview is not wasted on it. We call such a recommender an *interview-compatible instrument*.

**Five requirements.** An instrument fit for measuring conversational elicitation must simultaneously satisfy:

**R1 — full-profile SOTA.** On a full interaction history it must rank at the level of the strongest shallow collaborative-filtering baselines (EASE [33], RecVAE [31]), so that elicitation is measured against a recommender nobody can dismiss as weak [10].

**R2 — any-length cold-start fold-in.** It must ingest an evidence set of *any* cardinality, from 0 (fully cold) upward, without per-user retraining.

**R3 — channel-agnostic input.** Items, out-of-catalogue concepts, entities, and free text (and, for the continuous policy, direction tokens) must all fold through *one* interface as arbitrary embeddings, not a fixed categorical schema.

**R4 — uncertainty-native belief.** It must carry an explicit covariance over the user belief, not just a point estimate, so that a policy can act on *what it does not yet know*.

**R5 — monotonicity.** A truthful answer must never *lower* ranking quality; adding evidence should, in expectation, only help.

**The bifurcation observation.** The central empirical fact this chapter documents is that R1 and {R3, R4, R5} are *anti-correlated across the published literature*. The models that hit the full-profile bar—closed-form linear autoencoders (EASE [33], EDLAE [34]), amortised VAEs (Mult-VAE [19], RecVAE [31]), and training-free graph filters (GF-CF [30], BSPM [7], Turbo-CF [26])—are *item-only point estimators* on a one-hot indicator basis. The models with an uncertainty-native, multi-channel belief—soft-attribute Gaussian elicitation [4], item+attribute Thompson bandits [17], conjugate critique folds over a frozen factorisation [36], and per-item Bayesian NL elicitation [3]—do *not* report full-profile accuracy at the R1 bar. We call these the *accuracy corner* and the *belief corner*. No published system we are aware of occupies both (§4).

**Contributions.** This chapter is the instrument paper of the thesis. It does not claim a new elicitation policy (that is the companion method chapter); it claims the *recommender that makes policy measurement possible*. Concretely:

1. **The conjunction as the contribution.** We frame  $R1 \wedge R2 \wedge R3 \wedge R4 \wedge R5$  as a single requirement no published system meets, and give a six-family taxonomy and per-near-miss gap analysis (§3, §4) that locates exactly which requirement each candidate fails. We say we are *aware of none*, never that there *is* none.
2. **Three architecturally-inexpressible legs.** We isolate the three input channels that are impossible (not merely untuned) in the item-indicator basis of every R1-bar model: (i) signed/graded per-entity values, (ii) out-of-catalogue concepts, and (iii) continuous direction tokens. These are the narrow, defensible novelty legs of the channel-agnostic interface (§4, §5).
3. **The instrument.** A design for a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian precision-accumulator belief layer over its latent, into which every channel folds as a linear observation (§5). The frozen-tower+exact-conjugacy shape is the one mechanism with a principled monotonicity story [36], and it is what separates us from the co-trained, approximate-posterior near-miss of Büyük et al. [4].
4. **An acceptance battery as a reusable certificate.** We define three headline outcome gates—full-strength preserved (G0), posterior strictly shrinks per question (G1), and cold-start per-question NDCG rises on *full and tail* versus random (G2)—and, because these outcome gates admit several false positives (a sign-blind fold, a circular answer model, a decorative covariance), extend them into a full battery of per-build gates G0–G9 plus one-time certifications C1–C3 that add the semantic, artifact-control, and answer-model-transfer checks. We propose the battery as a portable certificate any interview-compatible recommender should pass (§2, §7).

We are careful throughout to distinguish what is *new* (the conjunction; the frozen-certified-tower wedge; the acceptance battery on a SOTA tower) from what is *pre-empted* and must be cited (attributes-as-directions plus a Bayesian fold [4]; unified item+attribute belief [17]; conjugate folds over a frozen factorisation [36]; permutation-invariant set-encoders with information-gain acquisition [24]; value-carrying set-input cold-start recommenders [20]; the non-degradation-of-information principle [12, 5]). “Uncertainty-native recommender” alone is a crowded 2024–25 area [6] and is *not* claimed as novel.

## 2 Requirements and gates

We now state the requirements and gates formally. Let the catalogue have  $n$  items and let the frozen tower expose a  $d$ -dimensional latent in which each item  $i$  has a factor row  $\mathbf{Q}_i \in \mathbb{R}^d$ . An *evidence set* after  $t$  questions is a set of observations  $\mathcal{E}_t = \{(\phi_j, y_j)\}_{j=1}^t$ , where  $\phi_j \in \mathbb{R}^d$  is the latent direction of the queried entity (an item factor, a concept centroid, an entity embedding, or a continuous direction token) and  $y_j \in \mathbb{R}$  is the (possibly graded, possibly signed) answer. The instrument maintains a belief  $b(\mathcal{E}_t) = (\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$  over the user latent  $\mathbf{u} \in \mathbb{R}^d$  and scores item  $i$  by a fixed read-out  $s_i(\boldsymbol{\mu}_t)$  of the base tower.

### Requirements (formal).

- R1.** With  $\mathcal{E}$  equal to a user’s full history,  $\text{NDCG@10}(s(\boldsymbol{\mu}(\mathcal{E})))$  ties the strongest shallow baseline on a shared harness (Def. 1).
- R2.**  $b(\mathcal{E}_t)$  is defined and computable for every  $t \geq 0$  with no gradient step at inference;  $t=0$  yields the prior belief (a non-personalised ranker).
- R3.** The update from  $\mathcal{E}_t$  to  $\mathcal{E}_{t+1}$  depends on the new observation only through the pair  $(\phi_{t+1}, y_{t+1}) \in \mathbb{R}^d \times \mathbb{R}$ ; the operator is identical for every channel, so any entity with an embedding is a legal observation—including entities absent from the catalogue.
- R4.**  $\boldsymbol{\Sigma}_t$  is an explicit, queryable covariance; a policy may read  $\boldsymbol{\Sigma}_t$  to choose the next question.
- R5.** For a truthful answer  $y_{t+1}$ ,  $\mathbb{E}[\text{NDCG@10}(s(\boldsymbol{\mu}_{t+1}))] \geq \text{NDCG@10}(s(\boldsymbol{\mu}_t))$ . We treat R5 as an *empirical* property to be demonstrated per step, not a theorem for a learned ranker (§8).

**Definition 1** (Ties SOTA). *A recommender ties a baseline on a metric if their scores are within the paired per-user bootstrap confidence interval on one shared harness (same users, items, split, metric depth). A number computed on a different metric/split (e.g. NDCG@10 on ML-25M vs. NDCG@100 on the ML-20M strong-generalisation split) is not a tie until bridged (§7.2).*

**Gates (acceptance battery).** The three headline gates below (G0/G1/G2) are *outcome* gates: they certify that the full-profile strength is preserved, that the posterior is usable, and that the cold-start curve rises. On their own they are insufficient (see below), so the instrument is accepted only against the full battery—per-build gates G0–G9 (every build, all cheap) and three one-time certifications C1–C3. Table 1 is the compact reference; the definitions follow.

*Per-build gates.*

- G0 — full strength & identity.** Evaluating on the belief mean with the full profile reproduces the frozen tower’s full-profile NDCG@10 (full *and* tail)—a paired-bootstrap tie, or bit-identity under the frozen-fold design—so the belief layer adds no full-profile regression (certifies R1 survives).
- G1a — posterior shrinks.** The posterior covariance shrinks in expectation over answers, with *refusals exempt* (a correct model does not shrink on a refusal).
- G1b — shrinkage is directional.** The shrinkage is at least twice as large along the queried  $\phi_{t+1}$  as elsewhere. G1a and G1b together certify R4: the covariance carries per-direction structure, not merely a global scale.
- G1c — belief scale calibrated.** The belief magnitude correlates with held-out NLL. This is a *diagnostic*, not an acceptance gate: it scores the posterior’s scale, on which no result in this chapter depends, and no comparator in the literature reports it.

- G2 — cold-start curve rises on full and tail.** Starting cold, per-question full and tail NDCG@10 must strictly *increase* from the first to the last turn (not merely fail to drop), judged by the *curve/endpoint at budget* rather than every-step significance; plus an out-of-envelope canary that a single true answer from cold helps on *every* channel. This is the operational form of R2∧R5, and it is a gate on *ingestion*: whether a better question *order* exists is a selection question, which this chapter deliberately does not adjudicate (§6.4).
- G3 — sign & intensity fidelity.** Negating every answer’s sign lowers NDCG (CI excluding zero, per channel); neutralising values to “indifferent” while freezing membership costs NDCG; and quality is monotone in stated intensity, so a binarising ablation must lose.
- G4 — refusals are inert.** A refusal burns the turn and moves the belief by nothing. In our design this is architectural rather than fitted—an unanswerable question is never folded—so the property holds by construction.
- G5 — concept/channel specificity.** Folding a concept ranks its member items above matched non-members (member-lift AUC > 0.8) *and* beats that concept’s own popularity projection (top principal component partialled out), with a dedicated channel at least matching a trained member-bag injection.
- G6 — existential controls.** Wrong-user, shuffled, and placebo-constant answers through the identical pipeline *gain nothing* relative to the true-answer arm (a control may legitimately fall *below* the no-answer intercept—a stranger’s taste is worse than no information, not equivalent to it), and duplicating an answer changes nothing (the belief responds to information, not cardinality).
- G8 —  $\Sigma$  load-bearing & set coherence.** Replacing  $\Sigma$  with an isotropic  $c\mathbf{I}$  *degrades* the acceptance-deciding number (else the R4 claim is withdrawn), and the belief and ranking are invariant to answer order and to conflicting re-asks.
- G9 — answer-ingestion isolation.** On a *fixed*, user-independent question set the curve still rises with  $t$  from answer content alone, separating instrument *ingestion* from question-selection *policy*.

*One-time certifications* (per instrument, not per build):

- C1 — published-number bridge.** The implementations snap to the published full-profile numbers on the canonical split (*met, with one disqualification*: EASE and RecVAE reproduce to the third decimal on the ML-20M Liang split and Mult-VAE near-snaps at  $-0.004$ , while Mult-DAE fails at  $-0.022$  and is dropped; §7.2).
- C2 — answer-model transfer (non-circularity).** The G2 gain survives under a *held-out behavioural* answer model plus injected label noise, at  $\geq$  a stated fraction of the self-model gain with a CI excluding zero, and a firewall assert guarantees that recommender-geometry answers appear nowhere in training or evaluation. This is the load-bearing certification: without it a G0–G2 pass is consistent with an instrument inverting its own simulator.
- C3 — human round-trip.** At least one study takes rendered questions  $\rightarrow$  graded human answers  $\rightarrow$  downstream NDCG. Deferred; carried as a project-wide external-validity obligation (§8).

**Why the outcome gates alone are insufficient.** G0/G1/G2 are outcome gates, and several broken instruments pass all three. A *sign-blind* fold that updates its belief from only the *direction* of the queried entity but never the answer *value* still reproduces the frozen tower (G0), still shrinks

$\Sigma$  along  $\phi$  per question (G1), and still beats a random-question control (G2)—because beating random certifies that a good question was *selected*, not that the answer was *ingested*; only the sign-flip test (G3) and the fixed-question ingestion isolation (G9) expose it. A *circular* answer model that generates answers from the instrument’s own latent geometry inflates the G2 curve with no transferable signal; only the held-out behavioural answer model and firewall of C2 close it. And a *decorative* covariance that is maintained but never enters the acceptance-deciding number passes G0–G2 trivially; only replacing  $\Sigma$  with an isotropic matrix and demanding a degradation (G8) shows whether the uncertainty is load-bearing. The battery’s semantic and artifact-control gates (G3–G9) and the transfer certification (C2) exist precisely to close these outcome-gate false positives.

We report *both* full and tail NDCG@10 for every gate (a project reporting rule: full-only or tail-only hides the effect, which is typically much larger on the tail than on the full catalogue). Of the battery, **G0 is passed on the certified tower** (§7, Table 5) and **every other per-build gate has been run** on a pre-certification snapshot of the same architecture and pass (Table 6). Results measured on that earlier snapshot are labelled **[RE-RUN QUEUED: ]** where the re-run on the certified tower is queued; **[TODO: ]** marks a gate never run at all.

### 3 Related work: a six-family taxonomy

We organise interview-relevant recommenders into six families and score each against R1–R5 (Table 2). The reading is consistent across families: no family is strong on both the accuracy axis (R1) and the belief-channel axis (R3–R5). Figure 1 plots the same fact quantitatively, with measured full-profile accuracy on one axis. Table 2 is the compact conceptual overview; the chapter’s anchor is the per-system *master table* (Table 3), which classifies every reasonable candidate against R1–R5 *and*, where one exists, prints its full/tail NDCG@10 on our shared ML-25M ruler—so that the bifurcation is not merely asserted from published claims but is visible in the one column of measured numbers.

**No dash is an unexplained omission.** Every unmeasured row carries one of three dispositions. (Systems with neither a measurable number nor a distinct capability profile—e.g. SLIM, strictly dominated by its closed-form successor EASE in every published comparison—are cited in the text but given no row.) *Run planned*: the graph filters and sequence models have public code and will be measured on this ruler in the next experimental wave. *Best-effort measured / planned*: for systems whose full-profile core is runnable but whose exact recipe is unpublished or benchmark-bound, we report a best-effort replication of that core with every substitution stated. Three are now *measured*, and they land weak: a Gaussian belief over an MF latent—the shared posterior core of Bıyık (no public implementation; their embedding is co-trained, so the substitution, if anything, flatters the original) and of ConTS’s Thompson posterior—measures 0.2019/0.1200 full/tail, far below the certified frontier (we no longer lean on the comparison with iALS, which is itself unbridged — §efsec:bridge); and the node recommender that is the Golbandi tree’s core measures 0.3065/0.1895. A trained Partial-VAE for EDDI and MF-seed reconstruction for RBMF/Functional-MF remain best-effort planned. That the belief corner’s runnable cores measure this far below the accuracy corner sharpens the bifurcation rather than softening it. *Not runnable by construction*: PEBOL maintains per-item Beta posteriors over an LLM-scored shortlist of  $\sim 100$  items and contains no full-catalogue ranker to run; GATE elicits but does not rank; zero-shot LLM ranking of an 18k catalogue is neither computationally nor methodologically meaningful at full-profile scale (and its collaborative weakness is already documented [13]); BCIE requires its authors’ knowledge graph,

and a port would substitute so much that the number would no longer be theirs—its conjugate-fold core is covered by the Bıyık-style best-effort arm. ICF is the one row whose full-profile core is *already* measured: its underlying model is Bayesian probabilistic MF, so its full-profile ceiling is the measured iALS row (0.2442).

**The measured-accuracy column falls as you enter the belief corner.** Read Table 3 down its measured-accuracy column. Every *competitive* number—a full/tail NDCG@10 at or near the frontier on our shared ML-25M ruler—sits in block (a) or (b): the item-only linear and amortised recommenders (RecVAE 0.3540, EASE 0.3476, EDLAE 0.3433). The belief corner, block (c), is either *blank* or *measurably weak*. Where its cores are runnable at all they land at the bottom of the column: the shared Gaussian-belief-over-MF core of Bıyık and ConTS at 0.2019/0.1200, the Golbandi node recommender at 0.3065/0.1895—and every one of these carries an ✗ on R1. The rest of the corner—EDDI, RBMF/Functional-MF, BCIE, UNICORN, PEBOL, GATE, the LLM-zero-shot CRS—has a blank accuracy column entirely. This is not an accident of our harness. Most CRS and elicitation papers never report full-catalogue top-N accuracy at all (they report elicitation RMSE, cold-start MRR/MAP on a shortlist, or critique metrics), and several *cannot produce it by construction*: a per-item Beta posterior (PEBOL) or the decision tree itself (Golbandi, whose node recommender we measure only as a best-effort core) does not induce a ranking over the whole catalogue, and an LLM CRS needs an API call per candidate to score one. Conversely, every system at the frontier is item-only and point-estimate—✗ on R3 and R4. The two blocks are disjoint: no row combines a frontier accuracy number with ✓s across R3–R5. The top-right cell of the gap map—all five requirements met *and* a competitive full-profile number—is occupied only by the target row, which is a design, not a result. That empty conjunction is the chapter’s thesis, and the master table is where it is impossible to miss.

**The accuracy corner (families ii–iii).** On the canonical ML-20M strong-generalisation split (10k validation / 10k test held-out users, NDCG@100), the full-profile frontier has been *flat since 2019–2020* and is held by shallow models, not deep nets: EASE [33] at NDCG@100 0.420, MultVAE [19] at 0.426, RecVAE [31] at 0.442 (top of this split), with EDLAE [34] in the same class; SLIM (0.401) and WMF/iALS (0.386) are the classic floors. Dacrema et al. [10] is the binding warning: most deeper “SOTA” gains vanish on this exact split. The graph-filter leaders (GF-CF [30], BSPM [7], Turbo-CF [26]) are the 2023–24 speed+accuracy front on Gowalla/Yelp/Amazon but do not report this ML-20M split, so they are admissible to our accuracy tier only if re-run on our harness. All of family (ii)/(iii) are item-only: their input is a one-hot (or set-indicator) interaction vector, which *cannot encode* a signed value, an out-of-catalogue concept, or a continuous direction—R3 is an architecture property here, not a tuning gap.

**The belief corner (family iv, and vi).** Bıyık et al. [4] maintain a multivariate-Gaussian belief over a user embedding and treat both item comparisons and “soft attributes” (e.g. “scarier”) as directions (concept-activation vectors) in the item space, selecting queries by expected value of information—the nearest thing to R2+R3+R4 together. ConTS [17] unifies items and attributes as Thompson-sampling arms with a posterior over a cold-start user. BCIE [36] folds critiques into a *frozen* factorisation by exact conjugate-Gaussian updates—the closest published analogue of our belief mechanism. PEBOL [3] keeps a per-item Beta posterior updated by LLM-extracted aspects and NLI scoring, reporting cold-start MRR@10 0.27 versus a 0.17 baseline. None of these reports full-profile accuracy at the EASE/RecVAE bar; each is a belief layer, not a certified tower.

**The set-encoder and cold-start-NP precedents (families i, iii).** EDDI/Partial-VAE [24] is the permutation-invariant set-encoder over an arbitrary observed subset with information-gain acquisition—exactly R2’s architecture, but demonstrated outside recommendation and without a full-profile CF result or an attribute/concept channel. TaNP [20] gives a neural-process, stochastic-process uncertainty over a value-carrying set input for cold-start—the R2+R4 shape, but not at full-profile SOTA nor channel-agnostic. These are cited as the mechanism precedents our design instantiates, not as rivals for the conjunction.

**The gap map.** Figure 1 places every system on two axes: *interview-compatibility* on the  $x$ -axis—a coarse ordinal capability count, the number of **R2–R5** marks a system satisfies in the master table ( $\checkmark=1$ ,  $\sim=0.5$ ; R1 is excluded because accuracy is the  $y$ -axis)—and *measured full-profile NDCG@10 on our ML-25M ruler* (R1) on the  $y$ -axis. The population is bimodal: the accuracy corner collapses onto  $x \leq 1.5$ , the interview-capable systems sit at  $x \geq 2$  with weak or non-existent full-profile accuracy, and no system bridges—the top-right, high compatibility *and* frontier accuracy, is empty.

## 4 Gap analysis

**The verdict.** No single published system satisfies all of  $\{\text{R1 SOTA-full-profile} \wedge \text{R2 any-length set} \wedge \text{R3 channel-agnostic via arbitrary embeddings} \wedge \text{R4 uncertainty-native} \wedge \text{R5 monotone}\}$ . The binding near-miss is Büyük et al. [4]: it fails R1 (a co-trained item space, not a frozen certified RecVAE-class tower, and no full-profile benchmark) and gives only an approximate, “generally not Gaussian” posterior; every other candidate fails R3 or R4 outright. **We are aware of none** that occupies the conjunction—and we state it exactly that way, never as “there is none.”

### Per-near-miss failure analysis.

- **Biyık et al. 2023** [4] — has R2 + R3 (items and soft attributes as directions) + R4 (Gaussian belief). *Fails R1*: the item space is co-trained with the belief model, not a certified full-profile CF tower, and accuracy is reported on their elicitation task, not the EASE/RecVAE bar. R3 is *attribute directions*, not arbitrary embeddings / open concepts / continuous query tokens. R5 is approximate (sampled EVOI over a non-Gaussian posterior). This is the central threat, and our wedge is exactly (a) frozen certified tower vs. co-trained, (b) exact conjugacy vs. approximate posterior, (c) arbitrary / open / continuous tokens vs. a fixed attribute set.
- **ConTS** [17] — R2 + R4 + a unified item/attribute action space. *Fails R3* (a fixed categorical attribute schema; no arbitrary embedding, no continuous or open token) and *R1* (a bandit over arms, not a full-profile encoder).
- **EDDI / Partial-VAE** [24] — R2 + R4 (information-gain) + set input. *Fails R1* (not a recommendation model / no CF SOTA) and *R3* (no attribute or concept channel as demonstrated).
- **RecVAE / EASE** [31, 33] — *R1 yes*. *Fail R3* (an item-only one-hot / indicator basis—cannot express a concept, a signed value, or a continuous direction) and *R4* (point estimate; RecVAE’s latent posterior is item-only and is not used as an actionable belief).
- **PEBOL** [3] — R4 (per-item Beta) + R3 (LLM aspects) + strong cold-start. *Fails R1* (no full-profile CF ranking) and its R4 is *per item*, not a shared user covariance; no monotonicity guarantee.



- **SASRec / BERT4Rec** [15, 35] — R2 (re-encode the sequence). *Fail R3, R4, R5* (item-only, order-sensitive, point estimate, non-monotone—a later token can reorder everything). Token input is trivially satisfied but is *not* value-carrying or mixed-type, so “interview = tokens” does not hold; and BERT4Rec’s fold-in advantage carries a replicability caveat [28].

**The three inexpressible legs.** Three input channels are *architecturally impossible* (not merely untuned) in the item-indicator basis of every R1-bar model: (i) graded/signed per-entity values (the basis has no sign and no magnitude—a “dislike” is not representable as a negative one-hot without changing the likelihood), (ii) out-of-catalogue concepts (there is no column for an entity absent from the catalogue), and (iii) arbitrary continuous direction tokens. Folding these as *linear observations in a frozen latent* is the one leg of the conjunction we find nowhere in the literature.

**Pre-empted—cited, never claimed as first.** We must credit, and do not claim priority over: attribute-answers-as-directions plus a Bayesian fold [4]; unified item+attribute belief and value-of-information [4, 17]; conjugate-Gaussian folds over a frozen factorisation [36]; permutation-invariant set-encoders with information-gain acquisition [24]; value-carrying set-input cold-start recommenders [20]; the non-degradation-of-information principle [12, 5]; and the now-crowded “uncertainty-native recommender” area [6]. Our defensible novelty is only the *conjunction* of all five requirements in one frozen certified tower, plus the three inexpressible legs and the G0–G2 protocol.

## 5 The instrument

The design follows directly from the gap analysis: use the shallow frozen tower that already holds the accuracy corner (R1, and the flat-since-2019 frontier means no deep-net detour is warranted), and add the one belief mechanism with a principled monotonicity story—exact linear-Gaussian conjugacy—over its frozen latent (R4, R5), with a channel-agnostic linear-observation interface (R3). The tower half of this design is built and certified (§7); the belief layer is specified here at design level and its numbers are **[TODO: ]**.

### 5.1 Frozen shallow-SOTA tower

The substrate is a certified shallow recommender—a RecVAE-class amortised encoder or a closed-form autoencoder [31, 33]—trained once on the full catalogue and then *frozen*. It exposes (a) item factor rows  $\mathbf{Q}_i \in \mathbb{R}^d$ , (b) a decoder read-out  $s_i(\mathbf{u}) = \mathbf{Q}_i^\top \mathbf{u} + \beta_i$  with a learned bias  $\beta_i$ , and (c) the encoder that maps a full interaction set to a latent. Freezing is the wedge against Bıyık [4] (co-trained) and the guarantee that G0 (full strength preserved) is a property of the substrate, not of the belief layer. We deliberately do *not* extend the linear basis with concept columns (that stays in the item-indicator world and fails R3), nor learn the uncertainty head (no monotonicity story, and retraining risks G0). The tower is now *selected and certified*: the graded-native set-encoder T2′, trained once in its pre-registered configuration, scores 0.3482/0.2462 full/tail on the held-out test cohort—a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar—and reproduces the frozen decoder exactly at the empty interview (§7). It is this checkpoint that the belief layer below is placed over.

## 5.2 Analytic conjugate-Gaussian belief layer

Over the frozen latent we place a Gaussian belief on the user vector,  $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ , and model each answer as a noisy linear read-out along the queried direction:

$$y_j = \boldsymbol{\phi}_j^\top \mathbf{u} + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \quad (1)$$

Because the observation is linear-Gaussian, the posterior after  $\mathcal{E}_t$  is Gaussian in closed form; maintaining it in *precision* form (a precision accumulator) makes the any-length fold-in a single running sum:

$$\boldsymbol{\Lambda}_t = \boldsymbol{\Lambda}_0 + \sum_{j=1}^t \sigma_j^{-2} \boldsymbol{\phi}_j \boldsymbol{\phi}_j^\top, \quad \boldsymbol{\Lambda}_t \boldsymbol{\mu}_t = \boldsymbol{\Lambda}_0 \boldsymbol{\mu}_0 + \sum_{j=1}^t \sigma_j^{-2} y_j \boldsymbol{\phi}_j, \quad \boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}_t^{-1}. \quad (2)$$

Equation (2) is exactly the mechanism BCIE [36] uses for critiques over a frozen factorisation; our contribution is doing it over a *certified full-profile SOTA* tower with channel-agnostic tokens, and certifying the G0–G2 protocol on it. Three properties fall out by construction: it is a permutation-invariant *set* operation defined for every  $t \geq 0$  (R2); each update adds a rank-1 term along  $\boldsymbol{\phi}_{t+1}$ , so  $\boldsymbol{\Lambda}_t$  only grows and  $\det \boldsymbol{\Sigma}_t$  only shrinks (G1); and conjugate updates are monotone in *expected utility* [12, 5]. The last is why R5 is plausible, but—because the tower’s dot-product ranker does not act optimally under the belief—per-question NDCG monotonicity is *not* automatic and must be shown empirically (§8).

**Status of the belief layer, stated precisely.** The layer is *built and fitted*, not hypothetical:  $\boldsymbol{\Lambda}_0, \boldsymbol{\mu}_0$  are taken from the tower, and the per-channel noise reduces to eight fitted scalars (a  $2 \times 3$  channel $\times$ confidence precision grid plus a prior scale and a variance floor), calibrated on real data with a pre-fit sign proof ( $G = 2291$  over 414,265 concept triples). It passes G1a, G1b, G4 and G9, misses G1c, and is partial on G8 (Table 6). Two limitations are load-bearing and we state them here rather than in a footnote. First, the covariance is maintained *decoupled* from the ranking path: the results in §6 are produced by folding answers to a point estimate, and  $\boldsymbol{\Sigma}$  is consulted only for question selection, where its advantage over random is +0.0036 at  $q=16$ . Second, the confidence axis of the precision grid is currently unexercised—MovieLens carries no within-channel confidence signal, so only the refusal band and a single answer tier are populated by real data. The honest claim is therefore that the instrument *provides* a correct, queryable, order-invariant belief, not that its covariance is doing heavy lifting in the numbers reported here; exploiting it is the companion chapter’s problem. **[RE-RUN QUEUED: refit the belief layer on the certified checkpoint and re-run Phase B]**

## 5.3 Channel-agnostic tokens as linear observations

Every channel supplies a direction  $\boldsymbol{\phi}_j \in \mathbb{R}^d$  and a value  $y_j \in \mathbb{R}$  to Eq. (2); the update is identical across channels (R3):

- **Items.**  $\boldsymbol{\phi}_j = \mathbf{Q}_i$  (the frozen item factor). A graded or signed answer sets  $y_j$ ; a dislike is a negative  $y_j$  along the same direction—representable here precisely because the observation model (1) has a sign and a magnitude, unlike a one-hot indicator.
- **Out-of-catalogue concepts.**  $\boldsymbol{\phi}_j$  is a concept direction in the latent (e.g. a centroid of member factors, or a text-adaptor image of the concept), folded exactly like an item—no catalogue column required.

- **Entities / free text.** An embedding is projected into the latent by a fixed adapter and folded identically.
- **Continuous direction tokens.** A policy may emit any  $\phi_j \in \mathbb{R}^d$  directly (the continuous-action interface the companion chapter uses); the belief update does not distinguish it from a catalogue entity.

The signed/graded, out-of-catalogue, and continuous legs are the three channels inexpressible in the R1-bar basis (§4).

**What is delivered, and what is still an interface rather than a result.** The signed/graded leg is delivered and measured (G3a/G3b, Figure 2). The concept leg is delivered for concepts *in a fixed vocabulary*: our implementation folds a concept by looking its direction up in a trained embedding table over the 1,031 genome tags, initialised from whitened member centroids. That is enough for R3’s out-of-catalogue clause in the narrow sense—no catalogue *column* is required, and a concept is not an item—but it is *not* yet the strong claim, which is that an arbitrary embedding, including one for an entity the model never saw in training, folds through the same interface. Everything downstream of the lookup already consumes a direction in  $\mathbb{R}^d$ , so the interface is arbitrary by construction; what is missing is evidence that a direction produced *outside* our own item geometry lands in the right place. We therefore state R3 as partially delivered and do not claim the continuous-token leg as demonstrated. **[TODO: the deciding experiment: fit one text adapter into the frozen latent, then fold English phrases that are neither genome tags nor titles, and show the ranking moves semantically (a “dinosaur” query surfacing the right films without the word appearing in any title), with a string-match control — the claim being *no per-concept training*, one adapter generalising to unseen phrases, which is the wedge against per-attribute probes [4]]**

## 6 The concept channel: derivation, lineage, and failure analysis

The out-of-catalogue concept leg is the most demanding of the three inexpressible channels—it is the one with no catalogue column, no interaction count, and no sign in the item-indicator basis—so we work it out in full. Three things must be made concrete: how a concept answer becomes an observation pair  $(\phi_c, y_c)$  for the belief update of Eq. (2); where each piece sits in the published record (so that we claim only the conjunction, never a component); and—because the honest history of this channel includes a self-inflicted regression—what a measured failure of our *own* evaluation harness taught us about which gate class the battery was missing. This section is written at derivation level; the concept-channel numbers below are measured on the internal ML-25M interview harness and are reported as such.

**Provenance convention for this section.** Because this section reports numbers from more than one build, we state once which stack each came from and mark the rest inline. *Accuracy* numbers (G0, the baseline table) are on the certified tower of §7. *Interview* numbers—every per-question curve in this section and the graded-input curve of Figure 2—are measured on a pre-certification snapshot of the same architecture, and the concept-channel curves additionally depend on which *fold operator* was trained: an earlier recipe whose concept capture was 1–6%, and the current one at roughly 25%. Results that stand on the current fold are reported plainly; results measured through the earlier fold are marked **[RE-RUN QUEUED: ]** and their re-runs are queued. Nothing in this section is a certified instrument curve, and we do not present it as one.

## 6.1 A signed, exposure-corrected affinity estimator

A concept answer must supply the value  $y_c$  that Eq. (1) reads along the concept direction. We want that value to be *signed* (a user can dislike “horror,” not merely fail to like it), *popularity-corrected* (a concept made of blockbusters must not read as universal affinity), *volume-corrected* (a heavy watcher must not read as liking everything), and *confidence-bearing* (a concept the user has plausibly never been exposed to is a *refusal*, not a dislike). Two estimators, an implicit and an explicit one, combine to meet this.

Let  $\mathcal{I}_u$  be the user’s watched set,  $p_i$  the catalogue marginal of item  $i$ , and let concept  $c$  have member set  $M_c$ . Write  $n_c = |\mathcal{I}_u \cap M_c|$  for the user’s watched members and  $e_c = |\mathcal{I}_u| \cdot \pi_c$  for the number *expected* under the user’s volume times the concept’s popularity  $\pi_c = \sum_{i \in M_c} p_i$ . The implicit *selection* estimator is the add- $\frac{1}{2}$  log watch-lift

$$\text{SEL}(u, c) = \log_2 \frac{n_c + 0.5}{e_c + 0.5} \approx \text{PMI}(u, c), \quad (3)$$

which is precisely the *pointwise mutual information* between the user and the concept: the denominator’s  $\pi_c$  divides out concept popularity, and forming  $e_c$  from the user’s own volume divides out user activity, so SEL is positive for genuine affinity, near zero for popularity-explained co-occurrence, and *negative for avoidance*. Its normalised form is the NPMI,  $\text{NPMI} = \text{PMI}/(-\log_2 \hat{p}_{uc}) \in [-1, 1]$ , which tames PMI’s small-count inflation and yields a bounded, comparable score. The explicit *value* estimator is a support-shrunk signed rating residual,

$$\text{VAL}(u, c) = \frac{n_c}{n_c + \lambda} \bar{r}_{uc}^{\text{res}}, \quad \lambda = 3, \quad \bar{r}_{uc}^{\text{res}} = \frac{1}{n_{c, i \in \mathcal{I}_u \cap M_c}} \sum (r_{ui} - \bar{r}_i), \quad (4)$$

the mean rating the user gives the concept’s members *above each item’s own mean*, shrunk toward zero by the Beta-style factor  $n_c/(n_c + \lambda)$  so that a one-film opinion counts for little. Both estimators are signed and combine into a four-band answer: a *graded like* (SEL/VAL band above the neutral zone), a *neutral* (indistinguishable-from-exposure), a *graded dislike* (band below), and a *refusal* issued exactly when the expected support falls below a threshold,  $e_c < \tau$ , i.e. when the user has plausibly never been exposed to the concept and no answer should be forced.

**Lineage.** Every piece is established prior art, which is the point of stating it explicitly. The log-lift (3) is PMI/NPMI, the standard popularity-corrected association score. Reading a *magnitude* of implicit consumption as evidence is Hu–Koren–Volinsky’s implicit-confidence template [14]—except that HKV lets the count re-enter *only as a loss confidence weight, never as a sign*, which is exactly the assumption our signed estimator breaks. The principled account of why a non-watch is not a dislike is ExpoMF [18]: it separates a latent *exposure* variable (was the user even aware of the item) from *preference*, so that non-consumption is modelled as “not exposed” *or* “exposed-and-disliked”—the generative gold standard our refusal band ( $e_c < \tau$ ) and popularity-corrected  $e_c$  approximate cheaply and in closed form. The unbiased-learning route to the same correction is attribute-level exposure propensity [29], which IPS-reweights implicit feedback by a per-attribute propensity; it is the heavier, variance-prone alternative to our moment estimator. Finally, calibrated recommendation [32] builds a user’s normalised genre *distribution*  $p(g \mid u)$ —the closest widely-used “concept profile,” but a distribution that sums to one and therefore encodes relative emphasis only: it *cannot express dislike*, which is precisely the expressive gap  $\text{SEL} < 0$  fills.

**Why the implicit channel, quantified.** On an informativeness decomposition (a per-concept linear fit of the oracle concept affinity onto each candidate answer,  $R^2$ ), the implicit watch-lift

dominates: SEL explains 0.376 of the affinity variance, the explicit rating residual VAL a further +0.042 (SEL+VAL 0.398), while an LLM-elicited ordinal answer explains only 0.017 (an internal decomposition)—implicit outperforms the explicit LLM answer by roughly  $22\times$ . Because SEL is built from raw watch counts and item marginals, it uses no recommender geometry, so this signal is *non-circular by construction* (it even beats a model-geometry projection of the same residuals). This is the evidence that the concept channel’s value is worth carrying signed rather than as a mere membership indicator; the decomposition is an informativeness measurement, not an interview curve, and is reported as such.

## 6.2 The fold operator: whitened directions and a gated fold-to-point

The direction  $\phi_c$  must point where the concept’s taste lives, not where popularity lives. A raw member centroid fails this: a shared popularity component (random concept pairs have cosine  $\approx 0.5$ ) masks the genre geometry. Removing it—centre the member factors and strip the top-1 principal component (the popularity axis), the all-but-the-top whitening of Mu and Viswanath [25]—turns the centroid into a usable direction: concept-membership discrimination through the frozen decoder rises from AUC 0.44 (raw, at chance) to 0.94 (whitened). We therefore take  $\phi_c$  to be the *whitened member centroid*, IDF-weighted by  $1/\log(1 + |M_c|)$  so broad genres do not dominate. Whitening is a fix, not a contribution, and is cited as such.

The operator that consumes  $(\phi_c, y_c)$  is where the design decision lives, and our own history is a clean ablation between two families. An *additive-union* operator scores each item by summing concept contributions on a popularity floor,  $s_i = \beta_i + \sum_c w_c \langle \mathbf{W}_d \mathbf{Q}_i, \phi_c \rangle$ . It has the right monotonicity floor (an empty interview scores exactly  $\beta_i$ ) but the wrong composition: it scores the *union* of directions, so correlated sibling concepts (“dark,” “gritty,” “noir”) re-add the same acclaim axis and over-count. Measured against a no-answer intercept of 0.1279 full NDCG@10, the additive channel peaks early (about +0.011 at two answers) and then *craters* as answers accumulate, reaching 0.0991 at eight answers—below the intercept. A *trained gated fold-to-point* operator fixes this by folding the answers into the latent itself and letting a frozen decoder apply the full collaborative (intersection) structure:

$$\mathbf{u} \leftarrow \mathbf{u} + \sigma(g([\mathbf{u}; \phi_c; y_c])) \odot h([\mathbf{u}; \phi_c; y_c]), \quad (5)$$

with the gate  $g$  and the delta MLP  $h$  *zero-initialised* (so at initialisation  $\mathbf{u}$  is unchanged and G0 survives) and the accumulated shift norm-capped per user. This operator is monotone in the number of answers to eight (concepts-only 0.2091/0.1222 full/tail at eight answers, up from the 0.1717 cold point) and, crucially, *redundancy-robust*: feeding it eight *correlated* top-SEL concepts still reaches 0.2219 full, at or above its four-answer level, where the additive-union operator degraded. **[RE-RUN QUEUED: operator-family magnitudes above are from the earlier fold recipe; the additive-vs-gated verdict is what carries forward, and the levels are re-measured with the fixed fold]** The gated fold is the concept operator we adopt; Eq. (5) is the concept-channel instantiation of the linear-observation fold, with the belief covariance maintained analytically as in §5.

**Lineage.** The skeleton—an attribute as a latent direction, folded by a Bayesian belief update over a frozen factor space—is well-trodden, and we claim none of it in isolation. Bıyık et al. [4] treat a soft attribute as a concept-activation-vector direction and fold signed answers into a Gaussian belief; M&Ms-VAE folds a rating-posterior and an attribute-posterior over one latent by a product-of-experts merge [1]; its successor adds an independent *signed* critique axis through learned positive/negative conditioning tokens, and reports that negative critiquing saturates after about five turns [2]—a concrete ceiling for any signed-attribute channel, consistent with our own front-loaded

concept curves. BK-VAE folds keyphrase-activation-vector directions by a Bayesian critiquing step [23]; the gate/MLP of Eq. (5) is a per-token generalisation of FiLM conditioning [27], which a single global affine cannot do because it would apply the same modulation to item and concept tokens alike. And the decision to *freeze* the tower rather than co-train a concept head is a direct reading of the cautionary result that a bolt-on critique head costs recommendation accuracy [22]. What none of these holds jointly—and what we are aware of no system holding—is the conjunction this channel occupies: *continuous, out-of-catalogue concept directions* (a whitened centroid, or a text-adaptor image, with no per-attribute probe to train), driven by a *signed, exposure-corrected implicit affinity* (Eqs. (3)–(4)), folded over a *frozen, certified full-profile SOTA tower*. The components are borrowed; only their conjunction is claimed.

### 6.3 Deployment currency: why a capability gate is not enough

A gate on what an operator *can* express must be paired with a gate on what the deployed system *does* express, and the concept channel is where the distinction bites. The fold of Eq. (5) is sign-capable: negating every answer moves NDCG by  $-0.201$  with a clean CI, so it passes G3’s sign-flip test outright. But that certifies the operator, not the interview. Driven down a realistic, user-independent question bank, most concept answers are weak positives, and an answer model that never emits a negative can accumulate them until the ranking drifts *below* the no-answer intercept while every capability gate still reads green.

We therefore evaluate the channel in *deployment currency*: a fixed, user-independent bank under a per-question budget where every question spends budget whether or not it is answered, scored at the endpoint against the intercept (the G9 ingestion-isolation setting). The acceptance form of the concept channel is not “can the operator express dislike” but “does the realizable interview stay above the no-answer intercept at budget.” This is the criterion the signed four-band estimator of §6.1 is built to satisfy, and §6.4 reports it.

### 6.4 Results: the signed retrain, and where the concept channel wins

The signed retrain of §6.3 has since landed, and we ran the two channels—concepts and items—through *one* deployment-currency harness (10,000 held-out cold users, per-question budget, every question spending budget whether answered or not, scored on full *and* tail NDCG@10 against the no-answer intercept  $0.1279/0.0192$ ). The results are reported here on the internal ML-25M interview harness; the certified instrument curves remain **[TODO: ]** (§7). They tell a consistent, and deliberately un-triumphal, story: the signed answer repairs the regression, but the concept channel’s value is *shaped*—it owns the opening of an interview and cedes the long tail of it to items—and its largest prize is one we do not yet capture.

**The signed answer turns the deployment decline into a gain.** Re-running the pre-registered triple gate on the *retrained* model (not merely the eval-only fold of §6.3) clears all three taste criteria. Signed answers beat the positive-only clip by  $+0.0104$  full NDCG@10 at the sixteenth question (CI  $[+0.0075, +0.0134]$ ): the clip-up curve sinks below the intercept by the eighth question and stays there ( $0.1268$  at  $q=16$ ), while the signed curve holds above it ( $0.1372$ ). The counterfeit control—an answer built only from user volume and concept popularity—reproduces *none* of the gain (it declines,  $-0.045$  at  $q=16$ ), so the signal is taste, not a volume $\times$ popularity artifact; and permuting the answer values across concepts collapses the gain ( $+0.0335$  CI-excluding-zero relative to the permutation), so it is the answer *values* that carry it, not fold cardinality. The two controls together establish what the channel is doing: the gain is carried by the answer *values*, and it is not



a restatement of how much the user has watched. [RE-RUN QUEUED: the triple-gate magnitudes above are measured through the earlier fold recipe; the three verdicts (recovery, counterfeit-clean, values-carry-it) are what the gate certifies, and the deltas are re-measured on the fixed fold]

**Both channels carry an interview.** The instrument’s claim on the elicitation side is deliberately modest, and it is worth separating from the claim we do *not* make. What the instrument chapter needs is that *both* channels fold and both move the ranking off the cold intercept—which is what makes the instrument usable for measuring policies over a mixed question bank. That much is unambiguous: an interview conducted entirely in out-of-catalogue concepts, a channel no R1-bar baseline can express at all, lifts a frontier-accuracy recommender substantially and monotonically above its 0.1279/0.0192 starting point, and so does an interview conducted entirely in items.

What we deliberately do *not* state here is a ranking between the two channels, and the reason is methodological rather than diplomatic. Any such comparison is only as meaningful as the question-selection strategy each channel is given, and pairing one hand-picked heuristic per channel—a popularity-ordered item bank against a polarization-ordered concept bank, say—measures the two heuristics at least as much as it measures the two channels. Every cross-channel comparison we have run so far has that defect, which is precisely why the earlier strategy×channel reading did not survive a change of fold. The comparison is therefore being redone under a single strategy-free criterion: the *best static question sequence* for NDCG, obtained by greedy maximisation at every step over an items-only bank, a concepts-only bank, and a combined bank of both, with the sequences constructed on the validation cohort and scored on the disjoint test cohort. Because the combined bank contains the items-only bank as a subset, it can only fall below items-only by overfitting the construction set—which makes the combined-versus- items gap a direct, interpretable readout of what the concept channel adds to an interview, rather than an artifact of whichever heuristics were paired.

The items-only arm has landed and is reported here as the reference line the other two banks will be read against: greedily selecting the best static item sequence from a 500-item bank on the validation cohort and scoring it on the disjoint test cohort gives full/tail NDCG@10 of 0.1405/0.0275 at one question, 0.1524/0.0387 at two, 0.1656/0.0445 at four, 0.1863/0.0538 at eight and 0.2098/0.0679 at sixteen, against the 0.1279/0.0192 intercept. The answerability figure is the striking one: across a sixteen-question interview a user answers on average only 3.6 of the item questions, the rest being items they have not seen. That is the structural pressure the concept channel exists to relieve, and it is why the combined bank is the interesting arm rather than a formality. [RE-RUN QUEUED: concepts-only and combined best-static arms (running); the combined—items gap is the chapter’s concept-versus-item evidence and no cross-channel claim is made until it lands]

**On the adaptive headroom.** The earlier suite also reported a large gap between a privileged oracle selector and every realizable static arm, with a naive greedy selector recovering only a low-single-digit fraction of it. Those magnitudes were measured through the same superseded fold and are withdrawn pending the re-run ([RE-RUN QUEUED: oracle—static gap and greedy closure on the fixed fold]). The *direction* survives on evidence independent of the fold: a one-level adaptive probe on the fixed stack—branch on the first answer, then choose the second question per branch, selected and evaluated out-of-sample on disjoint halves—pays on the item channel with a CI excluding zero (+0.0015 full, CI [+0.0007, +0.0024]; +0.0016 tail, CI [+0.0008, +0.0024]), and is positive but not significant on the concept channel at one level. The effect is small because a one-level preview is not a policy; that is the point. Establishing that adaptivity pays at all, on an instrument whose ingestion is certified, is what this chapter owes the companion method chapter,

whose job is to close the gap—the decision-tree acquisition line of Golbandi et al. [11] on *this* instrument and harness. The instrument chapter’s job is to make the gap *measurable*; closing it is the method chapter’s.

**Training concepts into the tower is item-safe but buys nothing.** We evaluated two homes for the signed concept fold: a trained fold-to-point on the *frozen* tower (call it the frozen-fold), and concept tokens trained *into* the tower itself (the tokens-in-tower variant). Both preserve full-profile strength, as G0 demands. The frozen-fold inherits the tower’s G0 by construction (a zero-initialised fold is bit-identical on the full profile). The tokens-in-tower variant is retrained and therefore must re-earn G0, and it does: full/tail NDCG@10 0.3487/0.2471 on the test users, a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar (difference  $-0.0053$ , within the 0.0070 CI), with cold-start accuracy if anything marginally higher. So *training concepts into the tower costs essentially nothing on items*. It also, however, *gains* essentially nothing on concepts, and it is worse where it matters most. On per-answer concept curves the frozen-fold is higher at eight answers (0.1630/0.0639 versus 0.1566/0.0577), and in deployment currency the difference is decisive: the frozen-fold stays above the intercept through sixteen questions (0.1379 at  $q=16$ ), whereas the tokens-in-tower variant *craters* to 0.1052, below the no-answer intercept, as the long tail of the fixed bank floods it with weak-support concepts. We therefore carry the frozen-fold as the concept operator of record; the tokens-in-tower arm is item-safe but does not earn its place. The G0 comparison above is a genuine tie on the certified ruler; the concept curves that decide *between* the two homes were measured through the earlier fold, so the ranking of the two homes is re-checked on the fixed fold before it is final. **[RE-RUN QUEUED: frozen-fold vs. tokens-in-tower concept curves on the fixed fold]**

**The methods lesson, restated as a battery amendment.** The concept channel’s near-miss is the sharpest argument in this chapter for *why the battery has the shape it does*. A capability gate (G3’s sign-flip) passed on a fold that, as actually driven down a realistic question bank, never emitted a negative answer—the answer model, not the operator, was sign-blind, and the operator’s certified capability was vacuous in deployment. Only the deployment-currency evaluation—a fixed, user-independent bank under a per-question budget, scored at the endpoint (the G9 ingestion-isolation setting)—exposed it, because there the declining curve penalises an answer model that never exercises the negative half. This is a general failure class, not a concept-specific accident: *a gate on what an operator can do must be paired with a gate on what the deployed system does*. We record the deployment-currency endpoint as the concept channel’s acceptance form, and fold the signed-retrain numbers into the G3/G9 rows once the certified tower is trained (§7).

## 7 Experimental design

### 7.1 Baseline bank

The baseline bank (Table 4) is three tiers: classic floors that every curve must clear (Dacrema et al. [10] demands them), the full-profile SOTA bar that decides G0, and the elicitation-compatible rivals that decide G1/G2. Reported anchors are on the systems’ *own* splits and are not comparable to ours until bridged (§7.2); we print them only to show what “ties SOTA” will mean.

### 7.2 Evaluation protocol

The protocol items below instantiate the headline outcome gates (the certification bridge is C1; G0 and G2 as named); they are the cheap, always-run subset. The remaining per-build gates (G3–G9)



and the load-bearing answer-model-transfer certification (C2), which close the outcome-gate false positives, are specified in full in the acceptance battery (Table 1) and its design-side document, and each headline result is reported against them.

**(i) Certification bridge (C1).** Our ML-25M ruler and the published bar are on *different* metrics and splits, and we do not conflate them. Concretely, our paper-config RecVAE realisation reaches full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M split—whereas the published EASE/RecVAE numbers are NDCG@100 on the ML-20M strong-generalisation split. These are not comparable, and no “ties SOTA” claim is made until the bridge is run. The bridge has two steps: (1) *reproduce* the published EASE and Mult-VAE/RecVAE numbers on the ML-20M Liang split (NDCG@100, Recall@20/50) in our own code, snapping to the reported values exactly [33, 19, 31]; (2) *port* the certified tower to the ML-25M harness and report full/tail NDCG@10 there. We keep ML-20M in the loop despite being older because the Liang 2018 split is a protocol lock-in the whole full-profile literature reports on; ML-25M is the newer corpus and, crucially, carries the genome tags the concept channel needs, so both datasets are required—one for comparability, one for the channels.

The first step is complete. On the ML-20M Liang strong-generalisation split our split statistics reproduce the published counts exactly (9,990,682 events / 116,677 train users / 20,108 items), and both anchor implementations **snap to their published values**: EASE at NDCG@100 0.4203 vs. the reported 0.420 [33], and RecVAE at 0.4425 vs. the reported 0.442, with Recall@20 (0.4144 vs. 0.414) and Recall@50 (0.5525 vs. 0.553) matching as well [31]. One closed-form and one neural reproduction to the third decimal certify that our split construction, metric implementation, and both implementation families are faithful to the published protocol—every ML-25M baseline number below therefore comes from certified code.

**Mult-VAE** reaches NDCG@100 0.4223 against the published 0.426 [19] (−0.0037; Recall@20 0.3928 vs. 0.395, Recall@50 0.5341 vs. 0.537)—a *near-snap*, close enough to admit as a bridged Tier-2 anchor but not a third-decimal reproduction, and we label it that way rather than round it into agreement. **Mult-DAE fails**: 0.3969 against the published 0.419 (−0.022, with Recall@20 and @50 short by a comparable margin). We do not know whether the gap is a training-recipe detail or something deeper, and rather than carry an uncertified implementation we **drop Mult-DAE** from the baseline bank entirely—it earns no row on the ML-25M ruler.

Consistency demands we apply that rule to ourselves, and doing so costs us something. Our iALS lands at 0.358 against the published WMF anchor of 0.386—a miss of 0.028, *larger* than the 0.022 that disqualified Mult-DAE. It would be a double standard to drop one and keep the other merely because the kept row is convenient, so we state the rule and its consequence explicitly: **iALS is an uncertified floor, not a bridged anchor**, it is excluded from every claim that requires a certified comparator, and in particular the sentence “the belief corner’s core measures *below even iALS*” is downgraded—it compares one unbridged number against another and is withdrawn as evidence. Only EASE, RecVAE and (as a near-snap) Mult-VAE carry certified status on this ruler. The bifurcation argument must stand on those or not at all.

**(ii) G0 — full-profile on ML-25M, items-only.** On the baselines’ native input (items only), evaluate full and tail NDCG@10 on the full ML-25M catalogue with a paired per-user bootstrap CI, confirming the instrument’s belief mean reproduces the frozen tower and ties EASE/RecVAE on the shared harness. The item-only baselines are measured (Table 5).

**A properly-trained RecVAE edges the closed-form frontier, and that is the bar.** On our canonical ML-25M ruler our in-house paper-config RecVAE attains 0.3540 full / 0.2497 tail, the nominally strongest measured system—ahead of the closed-form leaders EASE (0.3476/0.2441) and EDLAE (0.3433/0.2395). The RecVAE-over-EASE margin is +0.0064 full, about 1.8 unpaired standard errors; we therefore state it as a *nominal* lead with a paired per-user bootstrap CI pending, not a settled win. All rows share the same split, cohort, tail definition, and NDCG@10 routine (verified by code inspection). The point of interest for the instrument is that on this ruler a properly-trained amortised VAE—not a closed-form model—sits at the top of the frontier; the instrument’s R1 obligation is precisely to *tie this bar*, reproducing RecVAE-class full/tail accuracy through the frozen tower and then preserving it under the belief layer at G0. We report the whole frontier rather than excluding the stronger baselines.

**G0 is passed: the interview tower ties the bar, and is exactly the frozen decoder when the interview is empty.** One certified training run of the graded-native set-encoder tower T2’—a single run of the pre-registered winning configuration, selected on validation and scored once on the held-out test cohort—reaches full/tail NDCG@10 **0.3482/0.2462** against the RecVAE bar’s 0.3540/0.2497. The difference is −0.0057 full with a paired per-user bootstrap CI of 0.0069, so the two *tie* in the sense of Definition 1: the tower pays no measurable full-profile price for being interviewable. Two properties of that number matter more than its size. First, it is on the *same* ruler as every other row of Table 5—same split, cohort, tail mask, and NDCG routine—so it is a tie in the strict sense, not a bridged one. Second, the G0 *identity* half holds exactly and not merely statistically: with an empty evidence set the tower emits a zero latent ( $|\mu_0| = 0$  to numerical precision) and its scores are a rank-identical image of the frozen decoder’s learned bias (Spearman 1.0000), which is what licenses reading every later interview curve as evidence about *ingestion* rather than about a shifted baseline. The cold intercept of every curve in this chapter, 0.1279/0.0192, is that same non-personalised read-out. Accuracy is therefore no longer the open question for this instrument; what remains open is whether the belief layer preserves the tie once it is placed over the frozen latent, which is **[RE-RUN QUEUED: G0 re-checked at the belief mean on the certified checkpoint]** — the belief layer is fitted and gated (§7.3), but on the earlier snapshot.

### 7.3 The battery, measured

The battery is proposed in §2 as a portable certificate, so it owes its own results. Table 6 reports every gate. Rows other than G0 and C1 are measured on a pre-certification snapshot of the tower architecture; the re-run on the certified checkpoint is queued.

**Once a bank is answerable, which question you ask matters much less than that it is answerable.** In the G2 curves, a *random* question drawn from the answerable bank beats popularity-ranked items at  $q=1, 2, 4, 8$  (e.g. 0.2358 vs 0.2230 full at  $q=8$ ) and is only overtaken at  $q=16$  (+0.0022, CI-clean). HELF and the  $\Sigma$ -greedy selector are likewise within noise of random until the largest budget, where  $\Sigma$ -greedy wins by +0.0036 (CI [0.0016, 0.0057]). This is not a contradiction of G9, whose random arm is drawn from the *whole catalogue* and is therefore mostly unanswerable (mean 1.4 of 16 questions answered, versus 13.5 for the popularity bank) and which the popularity bank beats by +0.1264: the two “random” controls are different objects, and we name them separately. The lesson is that once a bank is restricted to answerable questions, *which* of them you ask matters far less than that they are answerable—which is precisely why the selection problem is deferred to the companion chapter and why we decline to claim a selection result here.

(iii) **G2 — per-question curves.** Cold-start per-question NDCG@10 curves (full and tail) versus a random-question control. Items are foldable by all baselines; mixed channels (concepts, signed values, continuous tokens) are foldable only by our instrument, which is the point of the comparison—the rivals cap out at what an item-only interview can carry. Symmetric access is enforced where a baseline *can* fold a channel; where it structurally cannot, that is reported as an architecture limit, not hidden. The items-only G2 curves and the G1 shrinkage diagnostic are measured (§7.3); what remains is the mixed-channel comparison against baselines that can fold only items. **[RE-RUN QUEUED: items-only and mixed-channel G2 curves on the certified tower, with the rivals of Table 4 on the arms they can express]**

(iv) **Concept-as-pseudo-item ablation.** To pre-empt “just add genre columns,” we fold a concept into EASE/RecVAE as the centroid-of-members pseudo-item and compare against our concept channel. The expected finding is that a pseudo-item column is a lossy special case (it cannot carry sign, magnitude, or an out-of-catalogue direction), but we run it rather than assert it. **[TODO: Ablation table: concept-as-pseudo-item fold into EASE/RecVAE vs. our concept channel, full & tail.]**

**Metrics.** Full *and* tail NDCG@10 on ML-25M (the project reporting rule, scored with the decoder’s learned bias, not a log-count popularity floor), plus NDCG@100 (and Recall@20/50) on the ML-20M bridge split for comparability with the published anchors. Every headline delta carries a paired per-user bootstrap CI on one shared harness. The G0 and baseline cells are measured; the G1/G2 cells are **[TODO: ]** pending the belief layer.

## 8 Limitations and threats to validity

- **Metric non-comparability until the bridge is run.** Our frontier number 0.3540/0.2497 is NDCG@10 on ML-25M; the published bar is NDCG@100 on ML-20M. Until §7.2 snaps our code to the published numbers (done for EASE/RecVAE, §7.2), *no* “ties SOTA” claim across the two rulers is licensed—the ML-25M and ML-20M numbers live on different rulers.
- **Monotonicity is empirical, not a theorem.** Conjugate-Gaussian updates are monotone in expected utility [12, 5] *only in expectation*, and only if the ranker acts optimally under the belief—which a fixed dot-product / NDCG ranker does not. R5 (and G2’s rising curve) is therefore a *measured* property with a per-question CI, not a guarantee; a truthful answer can in principle still lower NDCG at a given step, and we report any such step rather than suppress it.
- **Simulator-based evaluation pending a human study.** G1/G2 are measured under a stated answer model (a simulator), which shares structure with the instrument’s own latent; this risks circularity (the answer, the belief, and the metric are all functions of the same space). We add answer-noise ablations and a simulator-alignment limitation to every gate; the battery discharges this through certification C2 (the G2 gain must survive a held-out behavioural answer model plus injected noise, with a geometry-answer firewall) and treats the  $\geq 1$  human round-trip study (C3) as the load-bearing external validation—not yet run.
- **BERT4Rec replicability.** The sequence baseline’s fold-in advantage is subject to a documented replicability caveat [28]; we tune it to the published protocol and flag any residual gap rather than treat a single run as authoritative.

- **Graph-filter admissibility.** GF-CF/BSPM/Turbo-CF [30, 7, 26] do not publish the ML-20M strong-generalisation NDCG; they enter Tier 2 only if re-run on our split, and are otherwise reported as an accuracy ceiling on other datasets, not as a bridged tie.
- **Absence of evidence.** The conjunction claim is “we are aware of none,” bounded by our survey; a set-encoder / deep-set recommender that hits RecVAE-class full-profile accuracy *and* takes value-carrying mixed-type tokens would partly pre-empt  $R1 \wedge R2 \wedge R3$ . Our searches found no such reported system, which is absence of evidence, not proof of absence.

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| Gate                                            | One-line criterion                                                                                          | Failure class it excludes                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <i>Per-build gates (every build; all cheap)</i> |                                                                                                             |                                                     |
| G0                                              | Belief mean reproduces the frozen tower’s full/tail NDCG@10 (tie or bit-identity).                          | Full-profile regression from the belief layer.      |
| G1                                              | $\Sigma$ shrinks in expectation over confident answers (refusals exempt), directionally, calibrated to NLL. | Unusable / uncalibrated posterior.                  |
| G2                                              | Cold full/tail curve strictly rises to budget; one true answer helps every channel.                         | Flat-model and out-of-envelope artifacts.           |
| G3                                              | Sign-flip lowers NDCG; value-neutralisation costs NDCG; quality monotone in intensity.                      | Sign-blind and intensity-blind folds.               |
| G4                                              | A refusal burns the turn and does not move the belief (architectural invariant).                            | Unanswerable questions silently scored as evidence. |
| G5                                              | Concept lifts member items ( $AUC > 0.8$ ) above its popularity projection; channel $\geq$ member-bag.      | Non-specific / popularity-parasite concepts.        |
| G6                                              | Wrong-user, shuffled, placebo, and duplicated answers gain nothing over the true-answer arm.                | Cardinality confound; existential leakage.          |
| G8                                              | Replacing $\Sigma$ with $cI$ degrades the deciding number at budget; order- and re-ask-invariant.           | Decorative $\Sigma$ ; order-dependence.             |
| G9                                              | On a fixed user-independent question set the curve still rises from answer content.                         | Taste-leak: selection masquerading as ingestion.    |
| <i>One-time certifications (per instrument)</i> |                                                                                                             |                                                     |
| C1                                              | Implementations snap to published numbers on the canonical split (met; Multi-DAE failed and was dropped).   | Uncertified ruler / metric mismatch.                |
| C2                                              | G2 gain survives a held-out behavioural answer model + noise; geometry-answer firewall.                     | Circular answer model (self-simulator inversion).   |
| C3                                              | $\geq 1$ study: rendered questions $\rightarrow$ graded human answers $\rightarrow$ NDCG (deferred).        | Simulator-only external-validity gap.               |

Table 1: **The instrument acceptance battery.** Per-build gates G0–G9 (there is no standalone G7; the answer-model-transfer check is folded into C2) run on every build; certifications C1–C3 are one-time per instrument. The three headline outcome gates are G0/G1/G2; the semantic and artifact-control gates G3–G9 and the transfer certification C2 close the false positives that pass the outcome gates alone (text). The full specification, with each gate’s testable form, is the design-side battery document.



| Family                               | Representatives                                      | R1       | R2       | R3       | R4       | R5       | One-line verdict                                                             |
|--------------------------------------|------------------------------------------------------|----------|----------|----------|----------|----------|------------------------------------------------------------------------------|
| (i) retrain / meta-learn per user    | MeLU, TaNP [20], CDRNP                               | ~        | ✓        | ✗        | ~        | ✗        | R2/R4 precedent; not full-profile SOTA, not channel-open.                    |
| (ii) fold-in linear / closed-form    | EASE [33], SLIM, GF-CF [30], BSPM [7], Turbo-CF [26] | ✓        | ✓        | ✗        | ✗        | ~        | The full-profile bar (G0); item-indicator basis $\Rightarrow$ R3 impossible. |
| (iii) amortised inference encoders   | Mult-VAE [19], RecVAE [31], EDLAE [34], EDDI [24]    | ✓        | ✓        | ~        | ~        | ✗        | The architecture we live in; VAEs item-only, EDDI right shape/wrong domain.  |
| (iv) Bayesian belief over item space | Bıyık [4], ConTS [17], BCIE [36], ICF [37]           | ✗        | ✓        | ~        | ✓        | ~        | The R4/R5 heart; belief layers, not full-profile towers.                     |
| (v) sequence models                  | SASRec [15], BERT4Rec [35]                           | ~        | ✓        | ✗        | ✗        | ✗        | Interview $\neq$ ordered session; tokens item-only, order-sensitive.         |
| (vi) LLM-as-recommender / CRS        | PEBOL [3], GATE [16], LLM-CRS [13]                   | ✗        | ✓        | ✓        | ~        | ✗        | Best R3 (open text); weakest R1 (weak collaborative signal).                 |
| <b>CASPER (target)</b>               | <b>this chapter</b>                                  | <b>✓</b> | <b>✓</b> | <b>✓</b> | <b>✓</b> | <b>✓</b> | <b>The conjunction; no published system occupies it.</b>                     |

Table 2: Six-family taxonomy scored against R1–R5. ✓ met, ~ partial, ✗ not met. Scores are our reading of each family’s *published* claims and are qualitative; the last row is a target, not a result. R1 and {R3,R4,R5} are anti-correlated: families (ii)/(iii) hold the accuracy corner, family (iv) the belief corner, and none spans both. Adapted from the landscape analysis (c2).

[illegible]

Table 3: **Master table: every reasonable system by R1–R5 and by measured full-profile accuracy on one shared ruler.**  $\checkmark$  met,  $\times$  not met (our reading of each system’s *published* claims, consistent with Table 2). *full/tail NDCG@10* is on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; tail = top-33%-train-mass head mask, 190 head items), scored with the decoder’s learned bias. The Blyk and ConTS rows share one measured posterior core (belief-MF, best-effort; substitutions per the no-dash paragraph), so both print 0.2019/0.1200; the Golbandi row prints its node-recommender core. Statuses are *measured* (run on this ruler), *queued* (**TODO**: ], run pending), *queued* (**TODO**: ] : runnable, implementation/run scheduled on this ruler; best-effort = *superseded* (a measured successor row subsumes it), or *not runnable* / *not portable* (impossible by construction; mechanism in the text). The final column is each system’s own published full-profile anchor on its own split/metric—not comparable to our ruler until bridged (§7.2); “n/r” where the literature reports none. The RecVAE row is our own paper-config training run on this ruler and is the current full-profile frontier: the “ours” block adds the graded native interview tower T9/ whose certified run ties that frontier

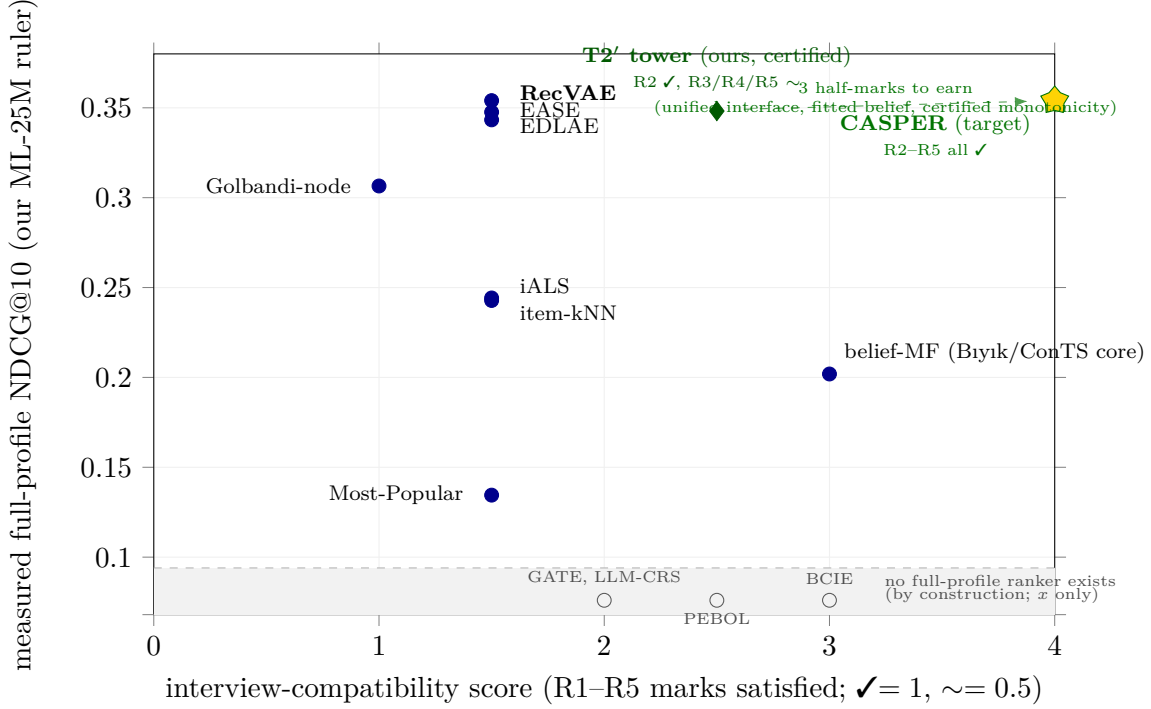


Figure 1: **The gap map (measured axes).** *y*-axis: measured full-profile NDCG@10 on our canonical ML-25M ruler. *x*-axis: an *interview-compatibility score*, defined as the number of **R2–R5** marks a system carries in the master table (Table 3), scoring ✓=1 and ~ = 0.5, maximum 4. *R1 is deliberately excluded from the score*: accuracy is the *y*-axis, and letting it also inflate *x* would blur the very trade-off the figure exists to show. The score is a *coarse ordinal capability count*, not a measurement, so the *x*-axis is qualitative even though the *y*-axis is measured. Filled markers are systems with a full-profile number on this ruler (belief-MF is the shared Biyık/ConTS posterior core; Golbandi-node is the tree’s best-effort node recommender). Hollow markers in the separated grey strip below the measured *y*-scale are systems that *cannot produce a full-profile number by construction* (no full-catalogue ranker, or not portable; see the master-table dispositions); only their *x* position is meaningful. Runnable-but-not-yet-measured systems (sequence models, graph filters, EDDI, TaNP) are deliberately *absent* from the figure — they carry **[TODO: ]** cells in Table 3 and join the plot when their numbers land. The star marks the instrument target (all five requirements at the frontier bar), still **[TODO: ]**; the green diamond is our *certified* tower T2’ at 0.3482—the first measured point that is simultaneously at the frontier bar and away from the item-only column. *The horizontal distance from the diamond to the star is not a record of failed gates*. T2’ scores 2.5 because it holds R2 outright and holds R3, R4 and R5 only *partially*: concepts fold, but through a separate trained operator rather than the single unified interface R3 asks for; the belief layer of §5 is specified but not yet fitted, so there is no queryable covariance (the same ~ we give RecVAE, and for the same reason); and per-question monotonicity is observed but not yet certified on this checkpoint. Three half-marks, each an unbuilt or uncertified component rather than a negative result—which is the honest reading of an instrument whose accuracy half is finished and whose belief half is not. The population is otherwise *bimodal*: the entire accuracy corner (Pop through RecVAE, 0.343–0.354 at the top) collapses onto  $x \leq 1.5$ —strong recommenders that can fold items and nothing else—while every system with genuine interview machinery sits at  $x \geq 2$  and either has no measurable full-profile accuracy at all (the grey strip) or, for the one measurable core, drops to 0.2019, below plain matrix factorisation (belief-MF at  $x=3$ ). Two clusters, no bridge: the corner is not expensive; it is unoccupied. The nearest published rival to the target is Biyık et al. [4] (highest compatibility of any system with an actionable belief), whose runnable core nonetheless measures far below the frontier bar.

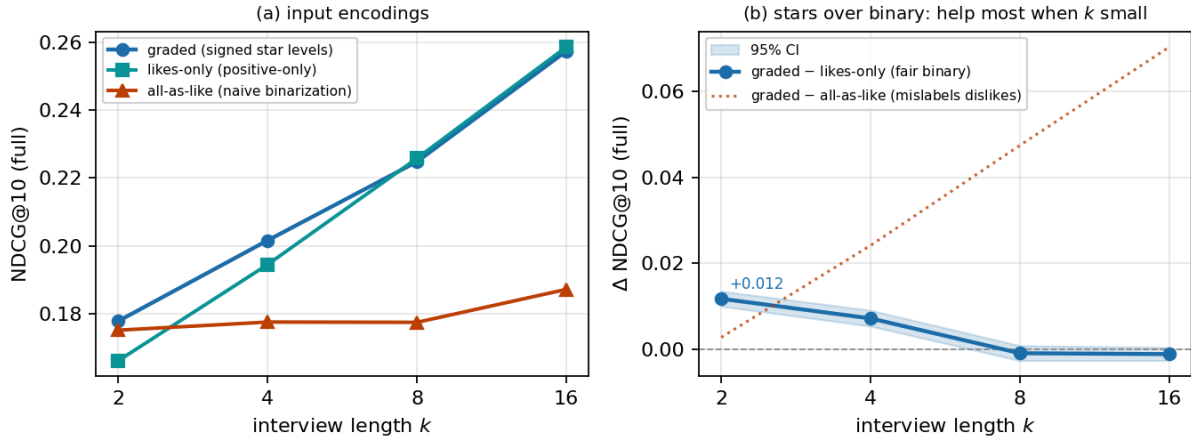


Figure 2: **Graded input helps over binarized input, most on the short interview (provisional; frozen graded-native tower, ML-25M interview harness, VAL, full NDCG@10 vs. interview length  $k$ ).** The same revealed set is fed three ways: *graded* (real signed star levels), *likes-only* (sub-threshold answers dropped—exactly what a positive-only tower such as RecVAE does at  $r>3.5$ ), and *all-as-like* (every answered item recorded as a 4-star like—naive implicit binarization). (b) Against the fair positive-only baseline, the graded premium is *CI-clean positive on the short interview*—+0.012 at  $k=2$  (CI [+0.010, +0.014]), +0.007 at  $k=4$ —and fades to a tie by  $k=8$  as collaborative signal fills in: the extra bits a star carries are most decisive when few items are known. The *mechanism* is that a signed answer encodes a dislike (a negative  $y_j$  the item-indicator basis cannot express); against the *all-as-like* control, which mislabels dislikes as likes, the apparent premium instead *grows* with  $k$  (dotted)—a control artifact (more revealed items  $\Rightarrow$  more dislikes corrupted), not value from gradation. Mirroring the star signs (flip) costs up to  $-0.29$ . Measured on the pre-certification snapshot of the tower architecture; **[RE-RUN QUEUED: graded-vs-binarized curve on the certified tower]**

| Baseline                                                     | Type                             | Reported anchor                            | Why the slot                                            |
|--------------------------------------------------------------|----------------------------------|--------------------------------------------|---------------------------------------------------------|
| <i>Tier 1 — classic floors (must clear)</i>                  |                                  |                                            |                                                         |
| Most-Popular                                                 | non-personalised                 | popb floor (our harness)                   | the “did we learn anything” line; G2 must beat it cold. |
| Item-kNN (cosine)                                            | neighbourhood                    | ML-20M NDCG@100<br>~ 0.36 class            | instant fold-in; Cremonesi-era strong-simple [8].       |
| PureSVD / iALS-WMF                                           | linear MF fold-in                | WMF ML-20M NDCG@100 0.386                  | cheap fold-in reference.                                |
| SLIM                                                         | sparse linear item-item          | ML-20M NDCG@100 0.401                      | direct EASE ancestor.                                   |
| <i>Tier 2 — full-profile SOTA (the R1 / G0 bar)</i>          |                                  |                                            |                                                         |
| EASE [33]                                                    | closed-form item-item            | NDCG@100 0.420                             | the bar to tie; item-only shows the R3 gap.             |
| RecVAE [31]                                                  | amortised VAE                    | NDCG@100 0.442                             | top of the ML-20M split; our in-house frontier.         |
| Mult-VAE [19]                                                | amortised VAE                    | NDCG@100 (ours 0.422) 0.426                | canonical amortised anchor; near-snap.                  |
| <i>Mult-DAE</i> (dropped)                                    | DAE                              | NDCG@100 (ours 0.397) 0.419                | snap <b>failed</b> -0.022; disqualified.                |
| EDLAE [34]                                                   | denoising higher-order EASE      | ≈ RecVAE class                             | confirms the flat frontier.                             |
| GF-CF / BSPM / Turbo-CF [30, 7, 26]                          | closed-form graph filter         | SOTA on Gowalla/Yelp (ML-20M not reported) | admit only if re-run on our split.                      |
| SASRec / BERT4Rec [15, 35]                                   | self-attentive sequence          | strong next-item (not this split)          | sequence fold-in; watch replicability [28].             |
| <i>Tier 3 — elicitation-compatible rivals (decide G1/G2)</i> |                                  |                                            |                                                         |
| EDDI / Partial-VAE [24]                                      | set-encoder + info-gain          | non-recsys (UCI/MNIST)                     | the R2 architecture; myopic info-gain baseline.         |
| Bıyık attributes [4]                                         | soft-Gaussian belief + EVOI      | own elicitation task                       | the central threat; frozen-vs-co-trained wedge.         |
| ConTS [17]                                                   | Thompson bandit, item+attr arms  | cold-start CRS metrics                     | unified-space + uncertainty; fixed schema.              |
| BCIE [36]                                                    | conjugate-Gaussian critique fold | critique metrics                           | exact-conjugacy-over-frozen precedent.                  |
| PEBOL [3]                                                    | Bayesian-opt NL-PE (LLM)         | MRR@10 0.27 vs 0.17                        | load-bearing PE baseline; per-item Beta.                |
| Golbandi tree [11]                                           | RMSE decision-tree interview     | Netflix RMSE                               | the static-tree interview comparator.                   |
| RBMF / Functional-MF [21, 38]                                | MF-embedding seeds               | elicitation RMSE                           | interview-seed line; cite-and-differentiate.            |

Table 4: Baseline bank. Anchors are on each system’s own split/metric and are *not* comparable to ours until the bridge (§7.2) is run; G0 is decided against EASE+RecVAE, G1/G2 against the Tier-3 rivals plus a random-question control. Any deep-net “SOTA” not snapping to Tier-2 numbers on our split is inadmissible [10].

| Method                              | full NDCG@10  | tail NDCG@10  | note                                            |
|-------------------------------------|---------------|---------------|-------------------------------------------------|
| Most-Popular                        | 0.1345        | 0.0226        | @100 0.1975; non-personalised floor             |
| belief-MF (Bıyık/ConTS core)        | 0.2019        | 0.1200        | @100 0.2886; best-effort posterior core         |
| Item-kNN (cosine)                   | 0.2428        | 0.1297        | @100 0.3200; neighbourhood fold-in              |
| iALS-WMF                            | 0.2442        | 0.1853        | @100 0.3704; MF fold-in                         |
| Golbandi node recommender           | 0.3065        | 0.1895        | @100 0.3958; best-effort tree core              |
| EDLAE [34]                          | 0.3433        | 0.2395        | @100 0.4330; val $p=0.4$                        |
| EASE [33]                           | 0.3476        | 0.2441        | @100 0.4375                                     |
| RecVAE (paper config; our run) [31] | <b>0.3540</b> | <b>0.2497</b> | @100 0.4537; nominal frontier                   |
| <i>Ours</i>                         |               |               |                                                 |
| Set-encoder tower T2' (certified)   | 0.3482        | 0.2462        | @100 0.4486; <b>G0 tie</b> (−0.0057, CI 0.0069) |

Table 5: G0 — full-profile, items-only, on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; seed 98765), NDCG@10 (full and tail); tail = top-33%-train-mass head mask (190 head items); scored with the decoder’s learned bias. The belief-MF and Golbandi rows are best-effort cores of the belief-corner systems (§3). All rows reuse the canonical harness verbatim, so they are on one ruler.

| Gate            | Criterion                                                         | Measured                                                                                                                                   | Verdict                 |
|-----------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| G0              | tie the bar; identity at $t=0$                                    | 0.3482/0.2462 vs 0.3540/0.2497, diff $-0.0057$ (CI 0.0069); $ \mu_0 =0$ , Spearman 1.0000 vs decoder bias                                  | <b>PASS</b>             |
| G1a             | $\text{tr } \Sigma$ shrinks over answers                          | 1881.9 $\rightarrow$ 1805.1 across $q=0 \dots 16$ , mono-tone                                                                              | pass                    |
| G1b             | directional, $\geq 2\times$                                       | along- $\phi$ 0.996 vs elsewhere 0.086 = <b>11.5<math>\times</math></b>                                                                    | pass                    |
| G1c             | $\rho(\text{belief-}\sigma, \text{NLL})$ (diagnostic)             | $\rho = 0.2393$                                                                                                                            | <i>reported</i>         |
| G2 <sup>†</sup> | cold curve rises, beats random                                    | $\Sigma$ -arm 0.1279 $\rightarrow$ 0.2724 full, 0.0192 $\rightarrow$ 0.1305 tail; rise +0.1445/ +0.1113                                    | <b>PASS<sup>†</sup></b> |
| G3a             | sign-flip lowers NDCG                                             | $-0.2985$ (CI $[-0.3047, -0.2922]$ ); intensity staircase $\rho = 0.927$                                                                   | <b>PASS</b>             |
| G3b             | value-neutralisation costs                                        | drop of 0.2869 (CI $[0.2817, 0.2922]$ ) full; tail 0.2999 $\rightarrow$ 0.0368                                                             | <b>PASS</b>             |
| G4              | refusals inert (architectural)                                    | unanswerable questions are never folded; $\alpha_{\text{refuse}}/\alpha_{\text{answer}} = 0.0007$                                          | <b>PASS</b>             |
| G5              | concept lifts members, beats its popularity projection            | whitened-direction membership AUC 0.94 (raw 0.44); cold fold clears the intercept 0.1308 $>$ 0.1279                                        | <b>PASS</b>             |
| G6              | wrong-user / shuffle / placebo / duplicate gain nothing           | shuffled +0.0371, placebo +0.0168 (both CI-clean), duplicate assert fired; wrong-user $-0.0986$ (a stranger's taste is worse than silence) | <b>PASS</b>             |
| G8              | $\Sigma$ beats isotropic $c\mathbf{I}$ at budget; order-invariant | +0.0089 at $q=16$ (CI $[0.0071, 0.0107]$ ), also wins $q=4, 8$ ; order deviation $5\times 10^{-7}$                                         | <b>PASS</b>             |
| G9              | fixed bank rises from answer content                              | rise +0.1218 (CI $[0.1184, 0.1252]$ ), mono-tone $\rho = 1.0$                                                                              | <b>PASS</b>             |
| C1              | snap to published numbers                                         | EASE +.0003, RecVAE +.0005, Mult-VAE $-.004$ ; <b>Mult-DAE <math>-.022</math> failed, dropped</b>                                          | <b>PASS<sup>†</sup></b> |
| C2              | held-out behavioural answer model                                 | <b>[TODO: ]</b>                                                                                                                            | —                       |
| C3              | human round-trip                                                  | deferred (§8)                                                                                                                              | —                       |

Table 6: **The acceptance battery, measured.** G1c is a diagnostic rather than an acceptance gate (§2) and is reported for completeness: the posterior is strongly directional (G1b, 11.5 $\times$  against a 2 $\times$  bar) while its scale tracks held-out NLL only weakly. G2 is measured on a bank filtered per user to the questions they can answer, so it isolates *ingestion* from question *availability*; a deployed static sequence faces the availability problem separately (§6). <sup>†</sup>C1 covers EASE, RecVAE and Mult-VAE (§7.2). Rows other than G0 and C1 are measured on a pre-certification snapshot of the same architecture.